

# Water and Radon Test Kit Descriptions

The Vermont Department of Health Laboratory (VDH lab) tests drinking and environmental water and radon in the air. Drinking water analyses are performed using EPA approved methods in accordance with the lab's NELAP accreditation.

## Ordering Test Kits

We offer multiple ways to purchase drinking water and radon test kits below.

**Online:** through our webstore <https://vdh.webapps.aidcv.t.com/water-test>

**Phone:** call 802-338-4724 or 800-660-9997 (toll-free in Vermont)

**Email:** download and email the [drinking water and radon test kit order form](#) to [AHS.VDHLabCustomerService@vermont.gov](mailto:AHS.VDHLabCustomerService@vermont.gov)

**Mail:** download and print the [order form](#)

**Fax:** download the [order form](#) and fax it to 802-338-4706



Our Drinking Water Sample Acceptance Policy and Instruction Sheets can be found on our website: [healthvermont.gov/lab/forms#water](http://healthvermont.gov/lab/forms#water). Samples not meeting our acceptance criteria will be rejected or qualified.

## Sending in Your Samples

Water samples should be collected the same day that they will be returned to the laboratory. Water samples can be submitted to the laboratory in 3 ways:

1. **Bring the sample to the Laboratory** at 359 South Park Drive, Colchester, VT from 7:45 am – 4:00 pm, Monday – Thursday. On Friday, samples are accepted from 7:45 am – 3:30 pm, with some exceptions noted.
2. **Bring the sample to the nearest Office of Local Health** on Monday – Thursday before the pickup time, typically in the morning. Pickup times are different for each office, and can be found here: [Drinking Water Drop-Off Program \(healthvermont.gov\)](#). Samples are not accepted on Fridays, holidays, or the day before a holiday.
3. **Mail the sample** – to ensure timely arrival, refer to sample collection instructions.
  - a) send USPS to PO Box 1125, Burlington, VT 05402-1125, or
  - b) send FedEx or UPS to 359 South Park Drive, Colchester, VT 05446

# Water and Radon Test Kit Descriptions (continued)

## Water Testing Packages

The packages below should cover most testing needs.

| Name                                                     | Description                                                                                                                                                                                                                                                                                                                                                     | Cost  |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Vermont Homeowner Testing Package</b>                 | Recommended for homeowners.<br>Kit A (bacteria, annually)<br>Kit RA (gross alpha, every 5 years)<br>Kit C (inorganic chemicals, every 5 years)                                                                                                                                                                                                                  | \$159 |
| <b>New Well or Spring Testing Package for Homeowners</b> | Designed to satisfy Vermont's requirements for new groundwater sources and most residential permit needs.<br>This bundle is composed of 4 kits: Kit A (bacteria), Kit RA (gross alpha radioactivity), Kit NW (selected metals and inorganics), and lead<br>Contains all analytes from tables A 11-5 and A 11-6 of the Wastewater and Potable Supply Rules 2023. | \$151 |

## Microbiological Testing

Samples must arrive at the laboratory within 30 hours of sample collection and should be kept cool. Test results are usually available 1 business day after the lab receives the sample. In cases of severe discoloration, Colisure testing may take 2 business days.

| Kit       | Description                                                                              | Cost |
|-----------|------------------------------------------------------------------------------------------|------|
| <b>A</b>  | Total coliform bacteria and <i>E. Coli</i> , presence/absence. Homeowner, Drinking Water | \$14 |
| <b>AA</b> | Total coliform bacteria and <i>E. Coli</i> , presence/absence. Regulated Water Systems   | \$14 |
| <b>NU</b> | Enumeration of total coliform and <i>E. Coli</i> in drinking water                       | \$15 |
| <b>AG</b> | <i>E. Coli</i> in Irrigation Water, Enumeration                                          | \$15 |
| <b>SW</b> | <i>E. Coli</i> in Recreational Water, Enumeration                                        | \$15 |

These tests are performed using Colisure or Colilert/Colilert-18 (SM 9223 B).

# Water and Radon Test Kit Descriptions (continued)

## Inorganic Chemical Testing

Most kits must be kept cold at no more than 42 °F (6 °C) and received within 45 hours of sample collection. See the chart on page 13 for specific timelines. Test results are usually available 18 calendar days after the lab receives the sample. Do not freeze.

|              |                              |                    |
|--------------|------------------------------|--------------------|
| <b>Kit C</b> | <b>Metals and Inorganics</b> | <b>Cost: \$100</b> |
|--------------|------------------------------|--------------------|

Recommended for homeowners every 5 years. This kit is a portion of the Vermont Homeowner Testing Package. Keep cold and return within 45 hours of collection.

EPA 200.7 Iron, Sodium, Total Hardness by Calculation

EPA 200.8 Arsenic, Copper, Lead (flush and first draw), Manganese, Uranium

EPA 300.0 Chloride, Fluoride, Nitrate, Nitrite

|               |                                                      |                   |
|---------------|------------------------------------------------------|-------------------|
| <b>Kit NW</b> | <b>Inorganic Portion of New Well Testing Package</b> | <b>Cost: \$80</b> |
|---------------|------------------------------------------------------|-------------------|

Keep cold and return within 45 hours of collection. This kit is a portion of the New Well or Spring Testing Package for Homeowners.

EPA 150.3 pH

EPA 200.7 Iron, Sodium

EPA 200.8 Arsenic, Manganese, Uranium

EPA 300.0 Chloride, Fluoride, Nitrate, Nitrite

|               |               |                   |
|---------------|---------------|-------------------|
| <b>Kit AN</b> | <b>Anions</b> | <b>Cost: \$50</b> |
|---------------|---------------|-------------------|

Keep cold and return within 45 hours.

EPA 300.0 Chloride, Fluoride, Nitrate, Nitrite, Sulfate

|                 |                            |                   |
|-----------------|----------------------------|-------------------|
| <b>Kit N3N2</b> | <b>Nitrate and Nitrite</b> | <b>Cost: \$24</b> |
|-----------------|----------------------------|-------------------|

Keep cold and return within 45 hours. EPA 300.0

# Water and Radon Test Kit Descriptions (continued)

## Inorganic Chemical Testing (continued)

Most kits must be kept cold at no more than 42 °F (6 °C) and received within 45 hours of sample collection. See the chart on page 13 for specific timelines. Test results are usually available 18 calendar days after the lab receives the sample. Do not freeze.

### **Kit IA**      **Trace Metals and Fluoride**      Cost: \$125

Regulated water systems. Return within 5 days of collection.

EPA 200.7      Iron

|           |           |           |            |          |
|-----------|-----------|-----------|------------|----------|
| EPA 200.8 | Antimony  | Chromium  | Mercury*   | Uranium  |
|           | Arsenic   | Cobalt    | Molybdenum | Vanadium |
|           | Barium    | Copper    | Nickel     | Zinc     |
|           | Beryllium | Lead      | Selenium   |          |
|           | Cadmium   | Manganese | Thallium   |          |

EPA 300.0      Fluoride

\*Mercury cannot be tested if the water is cloudy.

### **Kit IB**      **Lead and Copper**      Cost: \$20

Regulated water systems and homeowners. EPA 200.8.

### **Kit ID**      **Inorganic Chemicals**      Cost: \$90

Keep cold and return within 22 hours of collection.

**Not accepted after noon on Fridays.**

EPA 150.3      pH

EPA 200.7      Iron, Sodium

EPA 200.8      Arsenic, Manganese, Uranium

EPA 300.0      Chloride, Fluoride, Nitrate, Nitrite

SM 2150B      Odor

# Water and Radon Test Kit Descriptions (continued)

## A la carte offerings:

Samples requiring cooling must be received at 42°F (6 °C) or less, but not frozen.

Many of these tests need to be received quickly. See the chart on page 13 for specific timelines.

| Individual Test                                  | Method                               | Cost | Cooling required? |
|--------------------------------------------------|--------------------------------------|------|-------------------|
| Alkalinity                                       | SM 2320 B                            | \$12 | yes               |
| Antimony                                         | EPA 200.8                            | \$12 |                   |
| Arsenic                                          | EPA 200.8                            | \$12 |                   |
| Barium                                           | EPA 200.8                            | \$12 |                   |
| Beryllium                                        | EPA 200.8                            | \$12 |                   |
| Cadmium                                          | EPA 200.8                            | \$12 |                   |
| Chloride                                         | EPA 300.0                            | \$12 |                   |
| Chlorine residual, free                          | Hach 8021                            | \$12 |                   |
| Chlorine residual, total                         | Hach 8167                            | \$12 |                   |
| Chromium                                         | EPA 200.8                            | \$12 |                   |
| Conductivity                                     | SM 2510 B                            | \$12 | yes               |
| Cobalt                                           | EPA 200.8                            | \$12 |                   |
| Copper                                           | EPA 200.8                            | \$12 |                   |
| Fluoride                                         | EPA 300.0 or LACHAT<br>10-109-12-2-A | \$12 |                   |
| Hardness, total                                  | EPA 200.7                            | \$12 |                   |
| Iron                                             | EPA 200.7                            | \$12 |                   |
| Lead (flush or first draw)                       | EPA 200.8                            | \$12 |                   |
| Manganese                                        | EPA 200.8                            | \$12 |                   |
| Mercury (Cannot be tested if<br>water is cloudy) | EPA 200.8                            | \$25 |                   |
| Molybdenum                                       | EPA 200.8                            | \$12 |                   |
| Nickel                                           | EPA 200.8                            | \$12 |                   |

## Water and Radon Test Kit Descriptions (continued)

| Individual Test        | Method                | Cost | Cooling required? |
|------------------------|-----------------------|------|-------------------|
| Nitrate                | EPA 300.0 or P-EC-203 | \$12 | yes               |
| Nitrite                | EPA 300.0 or P-EC-203 | \$12 | yes               |
| Odor                   | SM 2150 B             | \$10 | yes               |
| pH                     | EPA 150.3             | \$10 |                   |
| Selenium               | EPA 200.8             | \$12 |                   |
| Sodium                 | EPA 200.7             | \$12 |                   |
| Sulfate                | EPA 300.0             | \$15 | yes               |
| Thallium               | EPA 200.8             | \$12 |                   |
| Total dissolved solids | SM 2540 C             | \$12 | yes               |
| Turbidity              | EPA 180.1             | \$12 | yes               |
| Uranium (Kit RU)       | EPA 200.8             | \$25 |                   |
| Vanadium               | EPA 200.8             | \$12 |                   |
| Zinc                   | EPA 200.8             | \$12 |                   |

# Water and Radon Test Kit Descriptions (continued)

## Radionuclide Testing

Samples are accepted Monday – Friday, 7:45 am – 3:30 pm.

Kit RC must be received within 48 hours of sample collection. Estimated turnaround times are listed below.

### Water test kits

| Kit       | Description                                           | Method     | Cost |
|-----------|-------------------------------------------------------|------------|------|
| <b>RA</b> | Gross Alpha<br>14 calendar day turnaround time        | EPA 00-02  | \$45 |
| <b>RC</b> | Radon-222 in Water<br>10 calendar day turnaround time | SM 7500-Rn | \$25 |
| <b>RU</b> | Uranium by mass<br>21 calendar day turnaround time    | EPA 200.8  | \$25 |

### Air test kits

Turnaround time 7 calendar days. Method EPA 402-R-92-004.

| Kit       | Description                | Exposure Time | Cost |
|-----------|----------------------------|---------------|------|
| <b>RF</b> | Radon in Air – Short Term  | 2 – 7 days    | \$50 |
| <b>RG</b> | Radon in Air – Medium Term | 1 – 3 months  | \$25 |
| <b>RH</b> | Radon in Air – Long Term   | 3 – 12 months | \$25 |

# Water and Radon Test Kit Descriptions (continued)

## Organic Chemical Testing

Kits must be kept cold at no more than 42 °F (6 °C) and must be received within 45 hours of sample collection. Do not freeze. Test results are usually available 21 calendar days after the lab receives the sample.

| <b>Kit OA</b>            | <b>Volatile Organic Compounds</b> | <b>Cost: \$120</b> |
|--------------------------|-----------------------------------|--------------------|
| EPA 524.2                |                                   |                    |
| Benzene                  | 1,1-dichloropropene               |                    |
| Bromobenzene             | cis-1,3-dichloropropene           |                    |
| Bromochloromethane       | trans-1,3-dichloropropene         |                    |
| Bromodichloromethane     | Ethylbenzene                      |                    |
| Bromoform                | Fluorotrichloromethane            |                    |
| Bromomethane             | Hexachlorobutadiene               |                    |
| n-butylbenzene           | Isopropylbenzene                  |                    |
| sec-butylbenzene         | P-isopropyltoluene                |                    |
| tert-butylbenzene        | Methylene chloride                |                    |
| Carbon tetrachloride     | Methyl tert-butyl ether           |                    |
| Chlorobenzene            | Naphthalene                       |                    |
| Chlorodibromomethane     | n-propylbenzene                   |                    |
| Chloroethane             | Styrene                           |                    |
| Chloroform               | 1,1,1,2-tetrachloroethane         |                    |
| Chloromethane            | 1,1,2,2-tetrachloroethane         |                    |
| 2-chlorotoluene          | Tetrachloroethylene               |                    |
| 4-chlorotoluene          | Toluene                           |                    |
| Dibromomethane           | 1,2,3-trichlorobenzene            |                    |
| 1,3-dichlorobenzene      | 1,2,4-trichlorobenzene            |                    |
| 1,2-dichlorobenzene      | 1,1,1-trichloroethane             |                    |
| 1,4-dichlorobenzene      | 1,1,2-trichloroethane             |                    |
| Dichlorodifluoromethane  | Trichloroethylene                 |                    |
| 1,1-dichloroethane       | 1,2,3-trichloropropane            |                    |
| 1,2-dichloroethane       | 1,2,3-trimethylbenzene            |                    |
| 1,1-dichloroethene       | 1,2,4-trimethylbenzene            |                    |
| cis-1,2-dichloroethene   | 1,3,5-trimethylbenzene            |                    |
| trans-1,2-dichloroethene | Vinyl chloride                    |                    |
| 1,2-dichloropropane      | m+p-xylene                        |                    |
| 1,3-dichloropropane      | o-xylene                          |                    |

# Water and Radon Test Kit Descriptions (continued)

## Disinfection By-Products

Kits must be kept cold at no more than 42 °F (6 °C) and must be received within 45 hours of sample collection. Test results are usually available 21 calendar days after the lab receives the sample.

Regulatory haloacetic acids and trihalomethane samples may need to be sampled at the same time and at the same location.

**These kits are not recommended for homeowners as they are primarily for regulatory testing.**

|               |                        |             |
|---------------|------------------------|-------------|
| <b>Kit OB</b> | <b>Trihalomethanes</b> | Cost: \$120 |
|---------------|------------------------|-------------|

Reported individually and as total trihalomethanes

|           |            |                      |
|-----------|------------|----------------------|
| EPA 524.2 | Chloroform | Dibromochloromethane |
|           | Bromoform  | Bromodichloromethane |

|               |                         |             |
|---------------|-------------------------|-------------|
| <b>Kit OK</b> | <b>Haloacetic Acids</b> | Cost: \$150 |
|---------------|-------------------------|-------------|

Reported individually and as HAA5

|           |                        |                               |
|-----------|------------------------|-------------------------------|
| EPA 552.2 | Bromoacetic Acid       | Dibromoacetic Acid            |
|           | Bromochloroacetic Acid | Trichloroacetic Acid          |
|           | Chloroacetic Acid      | Total Haloacetic Acids (HAA5) |
|           | Dichloroacetic Acid    |                               |

# Water and Radon Test Kit Descriptions (continued)

## Synthetic Organic Chemicals – SOC's

Kits must be kept cold at no more than 42 °F (6 °C) and must be received within 45 hours of sample collection. Test results are usually available 21 calendar days after the lab receives the sample.

These kits are not typically recommended for homeowners.

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| <b>Kit OE</b> | <b>Herbicides in Water</b> |                   |         | <b>Cost: \$200</b> |
|---------------|----------------------------|-------------------|---------|--------------------|
| EPA 515.4     | Dalapon                    | Dicamba           | Dinoseb |                    |
|               | Picloram                   | Pentachlorophenol | 2,4-D   |                    |
|               | 2,4,5-TP (silvex)          |                   |         |                    |

| <b>Kit OG</b> | <b>Carbamate Pesticides</b> |                    |                  | <b>Cost: \$100</b> |
|---------------|-----------------------------|--------------------|------------------|--------------------|
| EPA 531.2     | Aldicarb                    | Aldicarb sulfoxide | Carbaryl         |                    |
|               | Methiocarb                  | Oxamyl (Vydate)    | Aldicarb sulfone |                    |
|               | Baygon (Propoxur)           | Carbofuran         | Methomyl         |                    |
|               | 3-Hydroxycarbofuran         |                    |                  |                    |

| <b>Kit OH</b> | <b>EDB/DBCP/123-TCP</b>            | <b>Cost: \$80</b> |
|---------------|------------------------------------|-------------------|
| EPA 504.1     | 1,2-Dibromoethane (EDB)            |                   |
|               | 1,2-Dibromo-3-chloropropane (DBCP) |                   |
|               | 1,2,3-Trichloropropane (123-TCP)   |                   |

# Water and Radon Test Kit Descriptions (continued)

## Synthetic Organic Chemicals (continued) – SOC

Kits must be kept cold at no more than 42 °F (6 °C) and must be received within 45 hours of sample collection. Test results are usually available 21 calendar days after the lab receives the sample.

| Kit OL                                | Semivolatile Organic Compounds |                                        | Cost: \$250 |
|---------------------------------------|--------------------------------|----------------------------------------|-------------|
| Semivolatile chemicals and pesticides |                                |                                        |             |
| EPA 525.2                             | Acenaphthylene                 | di-n-Butyl phthalate                   |             |
|                                       | Alachlor                       | Endrin                                 |             |
|                                       | Aldrin                         | Fluorene                               |             |
|                                       | alpha-Chlordane                | gamma-BHC (lindane)                    |             |
|                                       | Atrazine                       | gamma-Chlordane                        |             |
|                                       | Anthracene                     | Heptachlor                             |             |
|                                       | Bis-(2-ethylhexyl) adipate     | Heptachlor epoxide                     |             |
|                                       | Bis(2-ethylhexyl) phthalate    | Hexachlorobenzene                      |             |
|                                       | Benz[a]anthracene              | 2,2',4,4',5,6-Hexachlorobiphenyl       |             |
|                                       | Benz[b]fluoranthene            | Hexachlorocyclopentadiene              |             |
|                                       | Benz[k]fluoranthene            | Indeno[1,2,3-cd]pyrene                 |             |
|                                       | Benzo[a]pyrene                 | Methoxychlor                           |             |
|                                       | Benzo[ghi]perylene             | Metolachlor                            |             |
|                                       | Butyl benzyl phthalate         | 2,2',3,3',4,5',6,6'-Octachlorobiphenyl |             |
|                                       | Dibenz[a,h]anthracene          | 2,2',3',4,6-Pentachlorobiphenyl        |             |
|                                       | Dieldrin                       | Pentachlorophenol                      |             |
|                                       | Diethyl phthalate              | Phenanthrene                           |             |
|                                       | 2,3-Dichlorobiphenyl           | Pyrene                                 |             |
|                                       | Dimethyl phthalate             | Simazine                               |             |
|                                       |                                | trans-Nonachlor                        |             |
|                                       |                                | 2,4,5-Trichlorobiphenyl                |             |
| EPA 505                               | Chlordane, total               | Toxaphene                              |             |

# Water and Radon Test Kit Descriptions (continued)

## Lead Testing for Schools

School lead kits must be ordered through the Vermont Tap Inventory Management System. Find more information here: [Testing for Lead in Drinking Water at Schools | Vermont Department of Health \(healthvermont.gov\)](#)

|               |                                    |                       |
|---------------|------------------------------------|-----------------------|
| <b>Kit SL</b> | <b>Lead (flush and first draw)</b> | Cost: \$12 per bottle |
|---------------|------------------------------------|-----------------------|

|                                                   |  |  |
|---------------------------------------------------|--|--|
| For schools performing Act 66 testing. EPA 200.8. |  |  |
|---------------------------------------------------|--|--|

## Child Care Water Testing

Child care licensure regulations now include public and private drinking water testing requirements. For more information about these regulations, contact the [Vermont Department for Children and Families](#) at 800-649-2642.

| Kit                   | Description                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------|
| <b>DC - Lead</b>      | Lead first draw or lead flush. Public and Private Water Systems                             |
| <b>DC - Coliform</b>  | Total coliform bacteria and <i>E. Coli</i> , presence/absence. Private Water Systems        |
| <b>DC - Inorganic</b> | Tests for Arsenic, Uranium, Manganese, Nitrate, Nitrite and Fluoride. Private Water Systems |

# Water and Radon Test Kit Descriptions (continued)

## When do my samples need to arrive at the lab?

Timelines given below are from the time of sample collection to receipt at the lab. Please make sure your samples arrive within the following timeframes. All kits not listed here should be received within 5 days of collection.

If you are planning to use the Drinking Water Drop-Off program at your local Health office, plan to take your water sample the morning you drop it off.

### 22 Hours

|        |                                   |
|--------|-----------------------------------|
| Kit ID | Chlorine residual, free and total |
| Odor   | pH                                |

### 30 Hours

Homeowner Testing Package  
Microbiological Testing (all kits)  
New Well Package

### 45 Hours

|          |        |           |
|----------|--------|-----------|
| Kit AN   | Kit OB | Kit OL    |
| Kit C    | Kit OK | Nitrate   |
| Kit N3N2 | Kit OE | Nitrite   |
| Kit NW   | Kit OG | Turbidity |
| Kit OA   | Kit OH |           |

### 48 Hours

|        |        |
|--------|--------|
| Kit RA | Kit RC |
|--------|--------|

# Water and Radon Test Kit Descriptions (continued)

## Additional Information

If you have questions about this guide, please call the Vermont Department of Health Laboratory at 802-338-4724 or 800-660-9997 (VT only).

Records relating to water and radon testing are public. Public records may be used for statistical purposes, and may be released upon request in writing, pursuant to Vermont access to public document law (1 VSA §315-320).

Instruction Sheets and Drinking Water Sample Acceptance Policy can be found on our website: [healthvermont.gov/lab/forms#water](http://healthvermont.gov/lab/forms#water).

Public water systems are regulated by the Vermont Department of Environmental Conservation. Learn more at [dec.vermont.gov/water/drinking-water](http://dec.vermont.gov/water/drinking-water).

Private wells and springs should be tested regularly. Learn more at [www.healthvermont.gov/water/testing](http://www.healthvermont.gov/water/testing).

For information on drinking water contaminants and treatment options, visit [www.healthvermont.gov/water-a-z](http://www.healthvermont.gov/water-a-z).

Find information on bacteria and radon in the environment at [www.healthvermont.gov/lab/environmental](http://www.healthvermont.gov/lab/environmental).

Find information on school and child care lead testing requirements at <http://www.HealthVermont.gov/lead-testing>